

Project Abstract

Group no. 3

Graduation project 1

CCAI-411

engine supervisor agent

University number	Student Name
2340507	Khaled Alotaibi
2342863	Abdulhadi Alfadli
2340550	Jad Felemban
2340394	Ghassan Alrefaei
2341614	Mohammed Alnejidi

When is preview worth acquiring? A dimensionless criterion, H/τ , for predictive thermal protection

A turbocharged engine protects its turbine housing only after damage begins: the control unit retards ignition once knock appears and enriches the mixture once exhaust gas is already too hot. Reinforcement learning could act earlier, given preview, meaning advance knowledge of the road gradient ahead.

Preview-based control is well established in automotive thermal management, yet preview is never free: it must be bought with mapping, connectivity, additional sensing and computation. The literature reports its benefit case by case and plant by plant, with no criterion that transfers between systems. An engineer deciding whether to acquire preview before building anything therefore has nothing general to consult.

This project proposes and tests a criterion governed by one dimensionless ratio: H/τ , the preview horizon divided by the thermal time constant of the protected part. If it holds, preview value falls on one curve in H/τ for any plant.

Completed pilot work calibrates a zero-dimensional cycle model and a three-node thermal network on 295.0 minutes of read-only OBD-II logging from the team's car; over 26 steady points, 30–75 kPa, modelled load matches the car to 1.4% with no fitted parameters. A Soft Actor-Critic agent will be trained and evaluated against a production baseline and a reactive policy under a fixed protocol, isolating preview by re-evaluating that policy blind. A battery plant tests transfer; non-overlap is itself a result.

The result is a design chart: knowing a component's time constant and the horizon available, an engineer can decide whether preview is worth buying. Manufacturers and researchers benefit. The physics runs live as a read-only supervisor estimating what the car cannot measure – protection before the damage, not after.

Application field: Automotive powertrain – engine thermal management, component protection, and on-board virtual sensing from read-only OBD-II data.

AI area: Deep reinforcement learning (Soft Actor-Critic) and preview-based predictive control.

Keywords: preview control; preview horizon; thermal time constant; dimensionless criterion; reinforcement learning; engine thermal protection; virtual sensing; OBD-II validation.